

Homework — 1.3.4 Web Technologies — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] 1 mark per error identified and corrected:

- The closing title tag is misspelled: `</titel>` should be `</title>`.
- The heading is never closed: the second `<h1>` should be the closing tag `</h1>`.

2 (a) [2] (a) `p` — an element/tag selector applies the style to every paragraph; (b) `.note` — a class selector; (c) `#warning` — an id selector, correct because an id is unique to one element on the page. Marks: 1 for (a); 1 for both (b) and (c) correct (the `.` and `#` prefixes are required).

2 (b) [1] Any one (1): the styling is defined once and applied to every page, giving a consistent appearance; one change to the CSS file updates the whole site, making maintenance far easier; it separates content (HTML) from presentation (CSS); the HTML files are smaller and the CSS file can be cached by the browser.

3. [3] In the order the rows are printed: **5, 3, 6, 1, 4, 2**. Correct sequence: 1 — a web crawler visits a web page; 2 — it records the page's keywords/metadata and notes its hyperlinks; 3 — it follows those hyperlinks to process further pages; 4 — the keywords are added to the index, mapped to the pages they occur on; 5 — a user's query is looked up in the index; 6 — matching pages are ranked (e.g. PageRank) and displayed. (2 correct rows per mark.)

4. [3] 1 mark per correct row:

- Row 1: **True**.
- Row 2: **True**.
- Row 3: **False** — the database of existing usernames is held on the **server**, which the browser cannot access, so this check must be done server-side (and client-side code could be bypassed anyway).

5 (a) [2] Each incoming link acts as a "vote" for the importance of the page it points to (1). A link from a high-ranking page passes on more rank / carries more weight than a link from a low-ranking page, so it increases the target page's rank by more (1).

5 (b) [1] The repeated recalculation stops when the rank values settle / no longer change significantly between iterations (1).

Part B (6 marks)

6. [2] A switch connects devices *within* a LAN and forwards frames to the intended device using **MAC** addresses (1). A router connects *different* networks (e.g. a LAN to the internet) and forwards packets between them using **IP** addresses (1).

7. [2] Symmetric encryption uses the *same* secret key to encrypt and decrypt, so the key itself must be transferred to the other party — and if it is intercepted in transit, all security is lost (1). Asymmetric encryption uses a public/private key pair: the public key can be shared openly because only the never-shared private key can decrypt, so no secret ever needs to be exchanged (1).

8. [2] A primary key is an attribute that uniquely identifies each record in a table (1). A foreign key is an attribute in one table (e.g. `CustomerID` in `Order`) that holds/references the primary key of another table, linking each order to its customer (1).

Total: 14 + 6 = 20 marks